

Human Computer Interaction (HCI)



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What is HCI?

***Design, Implementation and Evaluation* of interactive systems for HUMAN use.**

HCI encompasses the study of how people interact with computers and other digital devices, as well as the design of user interfaces and experiences to make technology more ***usable, efficient, and enjoyable*** for users.



What is HCI? - Human

- Human?? Or User
- The user is whoever is trying to get the job done using the technology
 - An individual User
 - Group of Users working together
 - Sequence of users in an organization, each dealing with some part of the task or process.



What is HCI? - Computer

- Computer??
- **any technology** ranging from the general desktop computer to a large-scale computer system, a process control system or an embedded system
- Any other???
- GUI
- Touch Screen and Gesture Recognition
- Voice User Interfaces
- VR
- Wearable Technology



What is HCI? - Interaction

- **Any communication** between a user and computer
- Can be direct or indirect
- Direct interaction involves a dialog with feedback and control throughout performance of the task
- Indirect interaction may involve batch processing or intelligent sensors controlling the environment



User Interfaces (UI's)

- Interaction between users and computers occurs at the user interface
- The golden principle in HCI is that “people should come first”.
- HCI includes understanding human capabilities, preferences, and behaviors in relation to technology, as well as designing interfaces and interaction techniques that accommodate diverse user needs and preferences.



Goal of HCI

- Reduce negative aspect such as frustration, annoyance..
- Enhance the positive aspect such as enjoyment, engagement...
- Or Enhance user experience by considering factors such as aesthetics, emotion and satisfaction.
- Make the product usable. Usable mean usability. Aims to design interfaces that are easy to learn, efficient to use and free from errors and confusions.



What is Usability?



“Extent to which a system, product or service can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use.”
- ISO 9241-11



What is Usability?

- Ease of learning
 - faster the second time and so on...
- Recall
 - remember how from one session to the next
- Productivity
 - perform tasks quickly and efficiently
- Minimal error rates
 - if they occur, good feedback so user can recover
- High user satisfaction
 - confident of success



HCI is Not about

- Making the interface look pretty, in fact interface should functionally effective too.
- Computer Science Programming
- Only about desktop computers (and that goes for computing as well!)
- Purely technical implementation: HCI practitioners focus on designing interfaces that meet user needs, preferences, and goals, rather than solely on technical specifications or constraints.



HCI is about

- Understanding the users
- Understanding users tasks
- Understanding the surrounding environment
- GUI requirements gathering and analysis
- Design prototype
- Evaluate the system
- **The goal of HCI** *“is to develop or improve the safety, utility, effectiveness, efficiency and usability of system that include computers.”*

(Interacting with computers, 1989, p3)



Why is usability important?

A Good user-interface can:

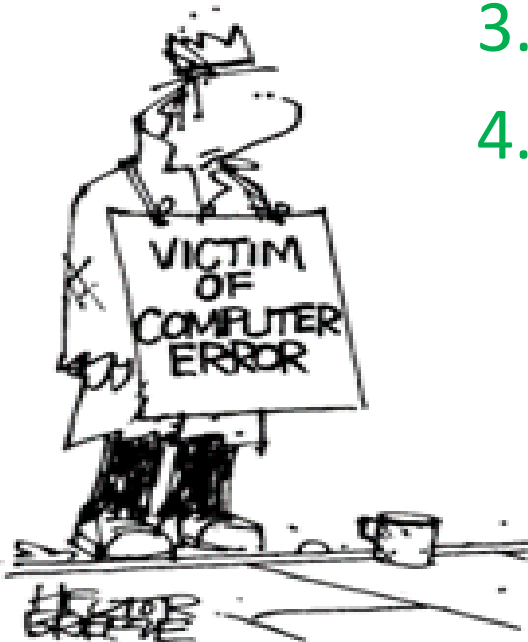
1. Earn a company billions.
2. Increase users loyalty.
3. Increase users trust.
4. Makes users happy :)



Why ..(cont.)

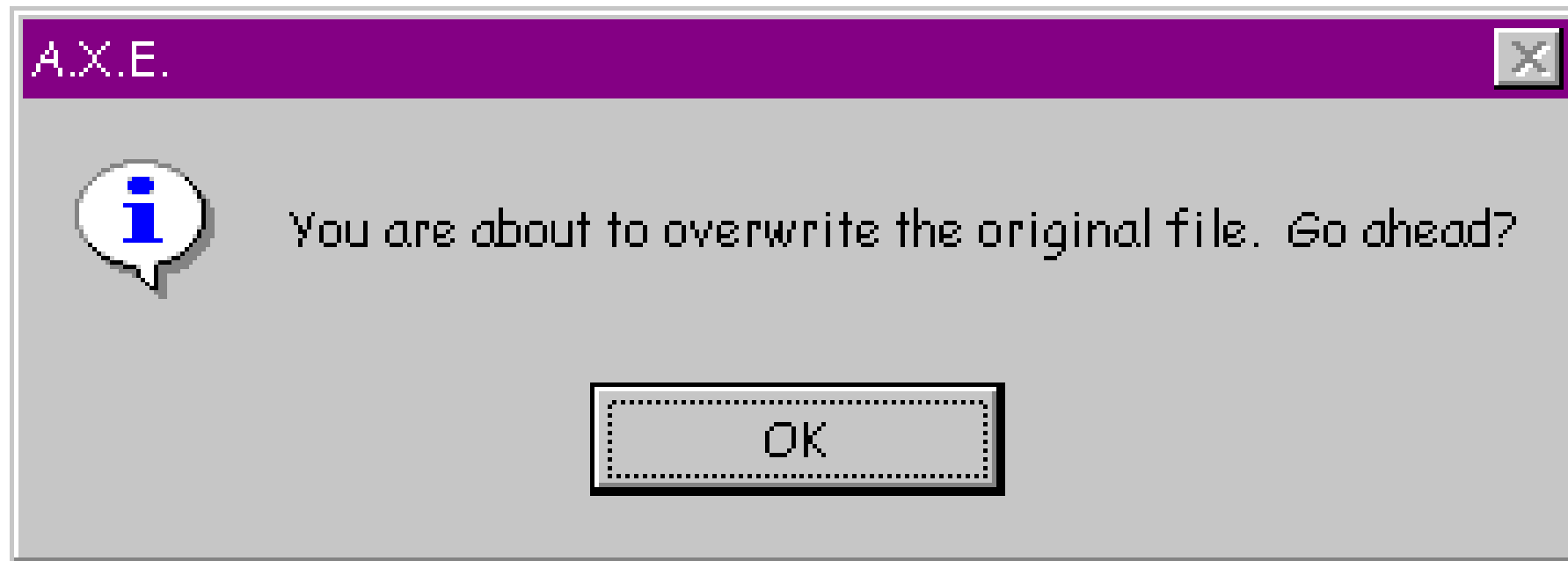
A Bad user-interface can:

1. be annoying, embarrassing, frustrating, and even deadly.
2. Increase mistakes in data entry and system operation.
3. Makes functions become completely inaccessible.
4. System failure because of user rejection.

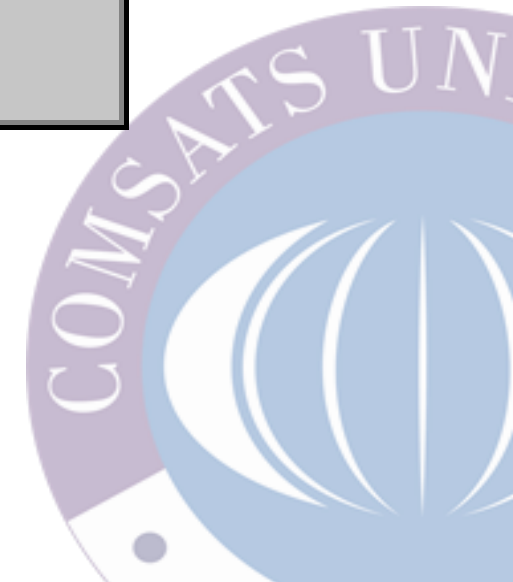


Examples of good and bad design





Do I have any choice?



Examples ..(cont.)

8) Age:

9) ☒ Female
☒ Male



Examples ..(cont.)



Examples ..(cont.)

▼ ONLINE BOOKING

☒ Flights ☐ Hotels

☒ Round Trip ☐ One Way [Multiple Dest.](#)

From: Jeddah

To: Jeddah

Depart: 03/12/2010

Return: 01/12/2010

☐ Flexible Dates

Adults (12+) 1

Children (2-12) 0

Infants (<2) 0

Class: Guest

SEARCH

1

2

3

4

5

6

7

8

9

► FLIGHT DETAILS

► MANAGE MY BOOKINGS

► CHECK-IN

The logo of COMSATS UNIVERSITY is partially visible on the right side of the slide. It features a circular emblem with a stylized globe or satellite dish design in the center, surrounded by the text "COMSATS UNIVERSITY" in a circular border.

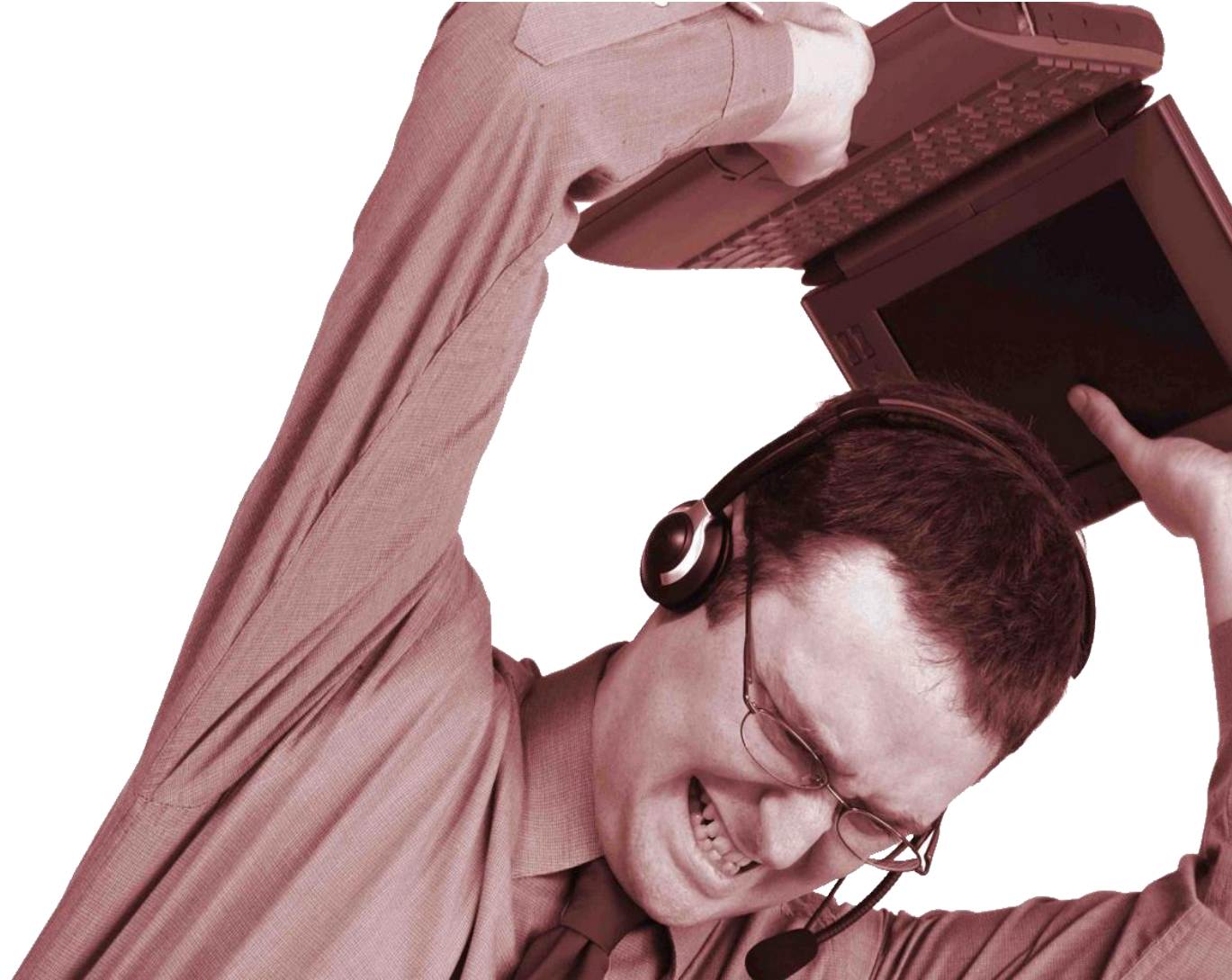
Examples ..(cont.)



Examples ..(cont.)



Is HCI really Important?



A real life example

A real example: a pilot shuts down the wrong engine and the plane crashes (as happened near Leicestershire, in England on the M1 motorway in 1989), this is obviously more serious. **47 died**



A real ..(cont.)

Example about: direct correlation between HCI and sales

- NYNEX: a telecommunication company in Italy
 - Purpose: to increase the performance of helpdesk office
 - Decided to improve the usability of the helpdesk operator interface
 - Reduced the process time 1 second per call
- Result: \$ 3,000,000 benefit / year**



Why study HCI?

- You will be building “real” systems
 - That other people will use
- UI’s major part of most systems
 - Often over 50%
 - 50% of effort rarely on UI!
- Bad UI’s cost
 - money (your product will be a flop)
 - lives (planes crash, reactors blow up)
- Interfaces are hard to get right
 - understanding of human capabilities will help
 - understanding principles of design will help



Who Builds Interfaces?

- A team of specialists (ideally)
 - graphic designers
 - interaction / interface designers
 - technical writers
 - marketers
 - test engineers
 - software engineers
 - customers

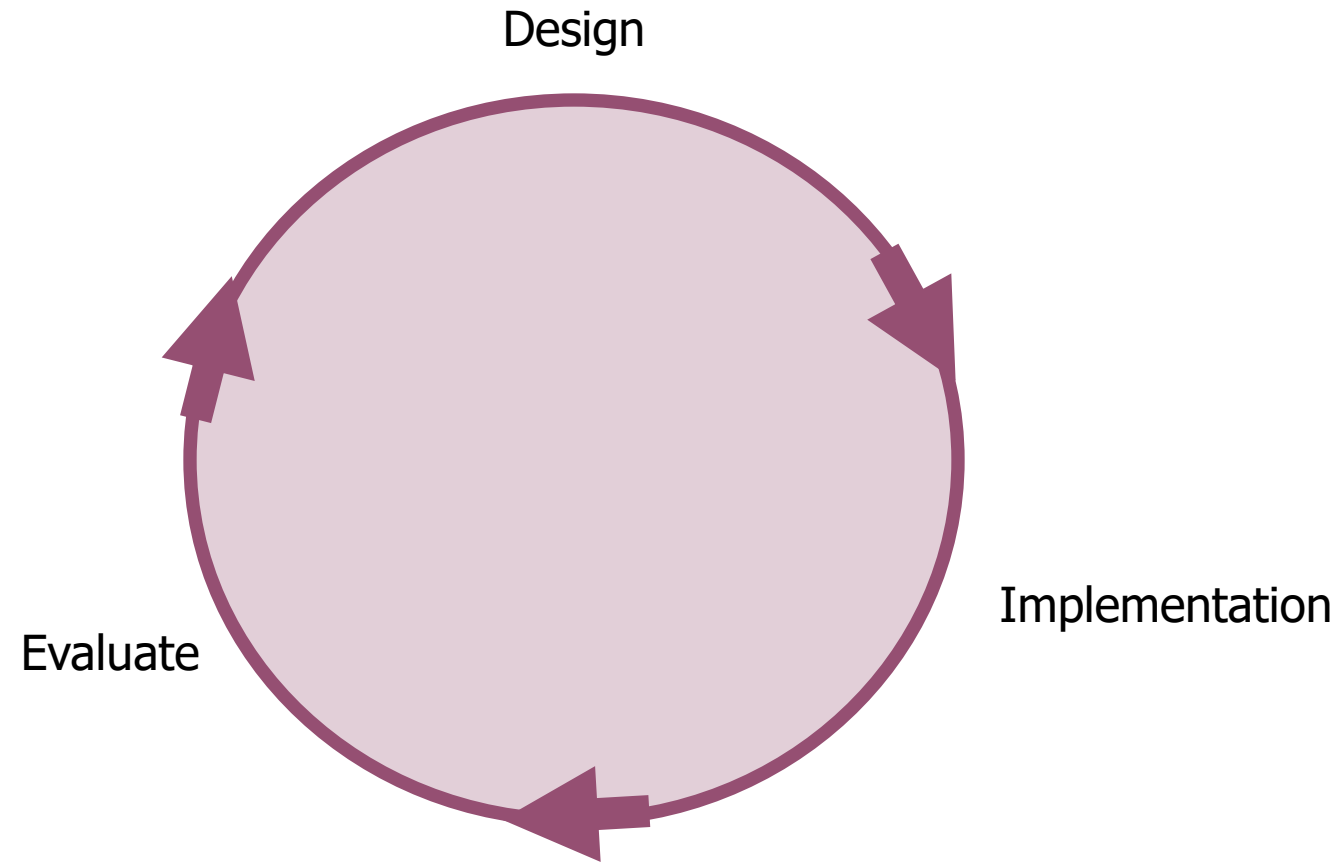


Foundations for building UI's

- Design Cycle
- Psychology of everyday things
- Understanding users and their tasks
 - Task centered design
- Design principles, usability heuristics ("Usability heuristics" refer to a set of broad guidelines or principles used to evaluate the usability of user interfaces)
- Designing with the user
 - User centered design
- Rapid prototyping
- Evaluation of the interface with users
 - Qualitative & quantitative
- Iteration



Design Cycle



A discipline concerned with the *design*, *implementation*, and *evaluation* of interactive computing systems for human use



Task analysis and design

- Observe existing work practices
- Create examples and scenarios of actual use of artifacts
- Try out new ideas with users before building anything if possible



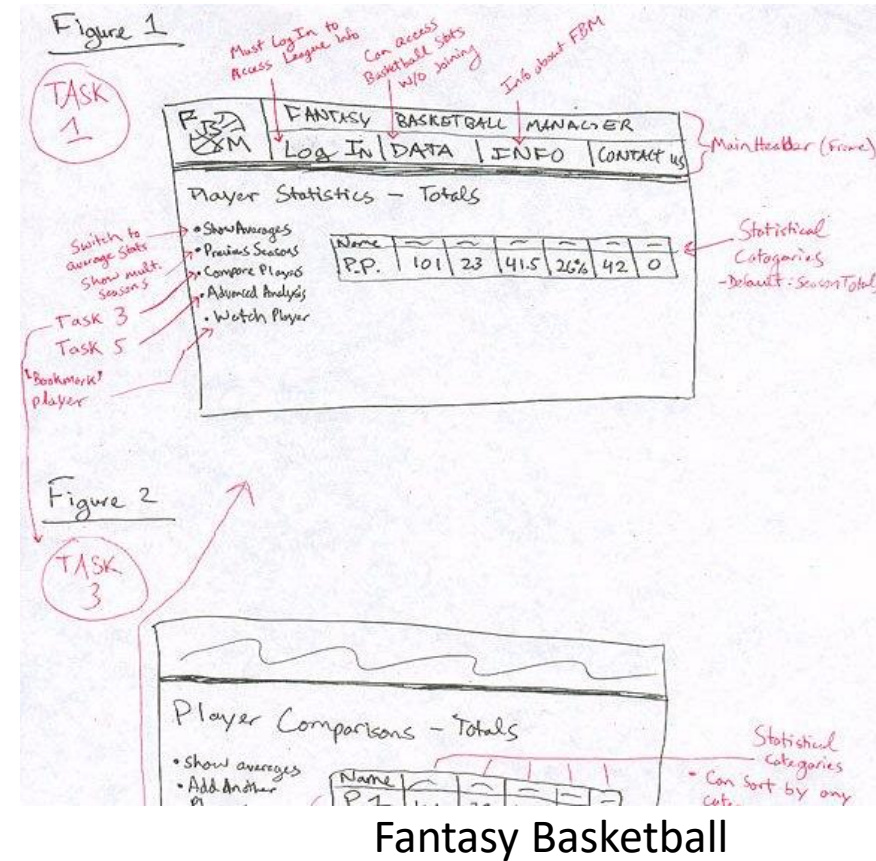
Rapid Prototyping

- Build mockup of design
- Low fidelity prototypes

- Paper sketches
- Wireframes
- Mockups
- Storyboarding

- High fidelity prototypes

- Somewhat working models
- High-fidelity prototypes are advanced prototypes that closely resemble the final product in terms of visual design, interactions, and functionality. They are typically used in later stages of the design process to refine and validate the user experience before final development



Evaluation

- “That’s cool!”, “I love it!” is NOT good enough
 - perception not always reality
 - conscious articulation not always behaviour
- Human behaviour & performance is complex
 - sometimes beyond analysis
 - individual differences
- Objective, quantitative, measures
- Qualitative techniques



Course Objectives

- Know what is meant by good design
- Have an understanding of human capabilities, design guidelines, models, and how to apply them to interface design
- Know and have applied some methods for
 - Design
 - Prototyping
 - Evaluation
- Know how to involve a user in the process
- Be familiar with a number of advanced topics in HCI
- Know how to communicate your work
- Have background to
 - Apply this to work in industry
 - Begin research in the area

